

A Research Principle for the AI Era

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Even when intrinsic value is the sole objective, the optimal strategy starts by investing entirely in external value.

The model below establishes this result for a sufficiently long research horizon.

Intrinsic value is the scientific contribution the researcher ultimately values. External value is support convertible into research funding, measured in the same units as expenditure.

Goal

Maximize expected total intrinsic value over n research cycles:

$$\max \mathbb{E} \left[\sum_{i=1}^n I_i \right].$$

Assumptions

Consider n research cycles, starting with a seed budget $c_1 = 1$. In cycle i , expenditure is c_i and the fraction allocated to intrinsic work is $p_i \in [0, 1]$. Write $\mathbb{E}_i$ for expectation conditional on the information available when p_i is chosen, including that choice.

1. Linear expected returns. For every decision history and allocation,

$$\mathbb{E}_i[I_i] = ap_i c_i, \quad \mathbb{E}_i[E_i] = w(1 - p_i)c_i, \quad c_{i+1} = d_i E_i.$$

The constants $a > 0$ and w are common to all cycles. The multipliers $d_i \geq 1$ are iid and bounded; each draw is independent of the decision information and current external return.

2. A growing reserve. For every decision history and allocation, the lowest possible external return is $b(1 - p_i)c_i$, with the same b across cycles. The funding balance obeys

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

For every possible outcome, require

$$B_i \geq \rho B_{i-1}, \quad 1 < \rho < b.$$

Research income must cover spending while growing the reserve.

3. Decision information. The researcher observes the current budget and balance and chooses p_i before the current returns and funding multiplier are drawn. Only b , w , and $\mathbb{E}[d_i]$ are known about their distributions.

The result

Theorem 1. *There is a finite integer $K \geq 1$, independent of the horizon n , such that whenever $n > K$, the unique optimal decision in each of the first $n - K$ cycles is to allocate all resources to external value:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

A numerical example

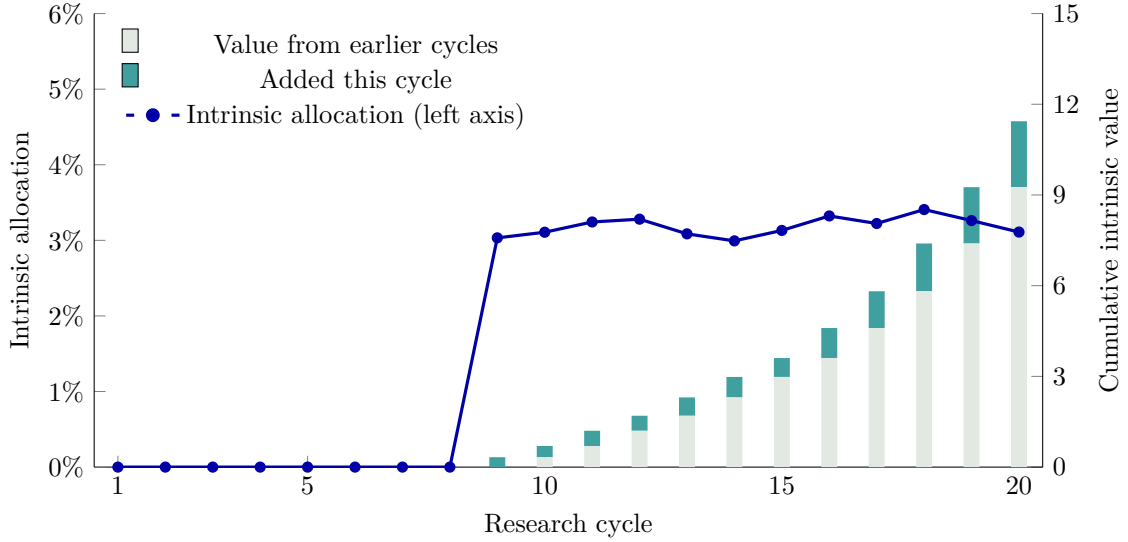


Figure 1: One optimal trajectory with iid random coefficients. All resources go to external value in the first eight cycles. The line shows the intrinsic allocation share. Each bar shows cumulative intrinsic value, with the current increment highlighted.

Use $I_i = a_i p_i c_i$ and $E_i = b_i (1 - p_i) c_i$, with $n = 20$, $\rho = 1.045$, and independent uniform draws

$$a_i \sim U[1.5, 2.5], \quad b_i \sim U[1.05, 1.15], \quad d_i \sim U[1, 1.2].$$

Here $K = 12$, so the first eight cycles are devoted entirely to external value. The figure shows one realization with random seed 20260928.

Proof

Every feasible state satisfies $0 \leq B \leq c$: this holds initially and is preserved because

$$0 \leq B' = B + E - c \leq E \leq d_i E = c'.$$

Write $u = (1 - p)c$ for external expenditure. The reserve constraint gives the feasible interval

$$\frac{c + (\rho - 1)B}{b} \leq u \leq c.$$

Divide the objective by a , giving expected stage reward $c - u$. Independence of the funding multiplier gives

$$v = w\mathbb{E}[d_i] \geq w \geq b > \rho > 1, \quad \mathbb{E}_i[c_{i+1}] = v(1 - p_i)c_i.$$

At every decision history with state (c, B) and m cycles remaining, the value is $V_m(c, B) = A_m c + D_m B$, with $A_0 = D_0 = 0$. To see this, define $S_m = vA_m + wD_m$. The Bellman objective is

$$(1 - D_{m-1})c + D_{m-1}B + (S_{m-1} - 1)u.$$

Its maximizing endpoint depends only on S_{m-1} , and both endpoints are linear in (c, B) , completing the induction.

When $S_{m-1} \leq 1$, choose the lower endpoint. Starting from $S_0 = D_0 = 0$, this gives

$$D_m = D_{m-1} + \frac{(\rho - 1)(S_{m-1} - 1)}{b} \leq 0$$

and

$$\begin{aligned} S_m &= \frac{v + w(\rho - 1)}{b} S_{m-1} + \frac{v(b - 1) - w(\rho - 1)}{b} \\ &\quad + (w - v)D_{m-1} \geq \rho S_{m-1} + b - \rho. \end{aligned}$$

Since $\rho > 1$ and $b > \rho$, this sequence crosses 1 in finite time. Let $K = \min\{m \geq 1 : S_m > 1\}$.

Once $S_{m-1} > 1$, the unique maximizing choice is $u = c$. Then $D_m = D_{m-1} \leq 0$ and

$$S_m = vS_{m-1} + (w - v)D_{m-1} > S_{m-1} > 1.$$

Thus every decision with more than K cycles remaining uniquely allocates everything to external value. These are precisely the first $n - K$ cycles. $\square$

Even when only intrinsic value matters, the optimal starting allocation is entirely external.